Methods and Determination of Quadrupole β_2 Deformation Parameters Across Experimental Techniques

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Abstract

This paper outlines the methodology for extracting nuclear quadrupole deformation parameters β_2 from reduced transition probabilities $B(E2)$, with a focus on Coulomb excitation as a clean electromagnetic probe. Theoretical foundations linking deformation to transition strength are presented, along with experimental techniques such as inverse kinematics, segmented detector arrays, and GOSIA simulations. A case study involving neutron-rich magnesium isotopes demonstrates the analysis pipeline and confirms enhanced collectivity with $\beta_2 \approx 0.5$ for $^{32\text{ and }30}$Mg, consistent with deformity expectations.

Contents

1 Introduction and Theory

This paper is a compact discussion about how the quadrupole deformation parameters β_2 are measured from reduced transition probabilities $B(E2)$ for a 0^+ to 2^+ transition. Specifically, the Coulomb excitation methodology is the main driver for this measurement, along with simulation tools such as GOSIA. This chapter provides a brief discussion of the theoretical background for defining β_2 and $B(E2)$, including their relationship and how to measure them. The following theoretical background entails starting with the shell model breakdown and how it leads to cases of nonspherical considerations, what the quadrupole formation is, the intrinsic quadrupole moment, and how this is all related to the transition probability $B(E2)$ and deformation parameters β_2.

1.1 Quadrupole Deformation Overview

The story of nuclear structure always starts with the shell model. Essentially, the shell model is an attempt to explain nuclear structure (how nucleons fill orbitals), in which protons and neutrons in isotopes can have specific orbital filling schemes (see Fig 1) that explain specific parities, total angular momenta, and other properties such as magnetic and quadrupole moments. Most importantly, the shell model explains how certain isotopes are more stable than others due to closed shells, or what we call ”magic numbers”. As shown in Figure 1, the shell model is governed by a Hamiltonian combining the harmonic oscillator, Woods-Saxon potential, and spin-orbit coupling as a mean field potential. This is important to note because the combination of all three is what provided correct approximations for achieving magic numbers [1]. A single-particle approximation is not an appropriate way to approach nuclear structure. One of the ways the shell model breaks down—leading to the presence of non-spherical nuclear shapes—is the ”island of inversion”; essentially, certain isotopes located near the neutron drip line no longer exhibit expected magic numbers (ex, ^{32}Mg see Fig. 2). This failure signals the emergence of strong correlations and collective behavior, leading us to think of nuclei as deformable many-body systems rather than spherical collections of independent particles.

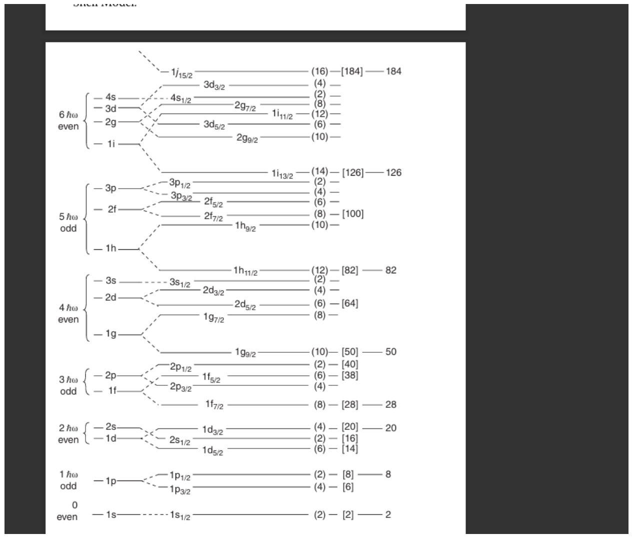


Figure 1: Example of a shell model spectroscopic chart based on how nucleons fill orbitals (harmonic oscillator + Woods-Saxon + spin/orbit). Image taken from [1].

When such deviations from spherical symmetry occur, collective motion becomes the dominant feature of nuclear structure. This includes both rotational and vibrational degrees of freedom. In particular, quadrupole deformation arises when the nuclear shape deviates from spherical symmetry, assuming either a prolate (rugby-ball) or an oblate (pancake-like) configuration.

The shape of the nuclear surface can be mathematically described using spherical harmonics. For quadrupole deformation, the nuclear surface is expanded as:

$$R(\theta) = R_0 \left(1 + \beta_2 Y_{20}(\theta)\right), \tag{1}$$

where:

- R_0 is the radius of a spherical nucleus,
- β_2 is the quadrupole deformation parameter,
- $Y_{20}(\theta) = \sqrt{\frac{5}{16\pi}}(3\cos^2\theta - 1)$ is a second-order Legendre polynomial (spherical harmonic).

The deformation parameter β_2 quantifies the degree and nature of the deformation. A positive β_2 implies a prolate shape, while a negative β_2 implies an oblate shape.

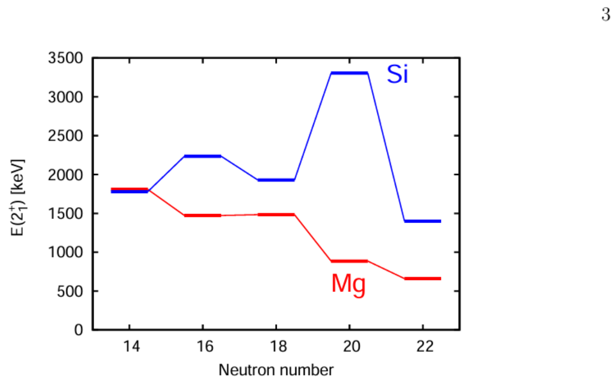


Figure 2: Excitation energies of the first 2^+ state for even-even Si (blue) and Mg (red) isotopes as a function of neutron number. The trend shows enhanced collectivity (lower $E(2^+)$) in neutron-rich Mg isotopes, consistent with quadrupole deformation and breakdown of the shell model near the so-called island of inversion. Image from [2].

1.2 Intrinsic Quadrupole Moment

One of the essential features, or "measurements," that arise from collective motion—or, more specifically, from non-spherical assumptions about nuclear shape—is the intrinsic quadrupole moment, Q_0. This quantity characterizes the degree to which the nuclear charge distribution deviates from spherical symmetry in the body-fixed (intrinsic) frame of a deformed nucleus.

For axially symmetric deformed nuclei, the intrinsic quadrupole moment is defined as:

$$Q_0 = \int \left(3z^2 - r^2\right) \rho(\vec{r})\, d^3r \tag{2}$$

Where $\rho(\vec{r})$ is the charge density distribution, r is the radial distance from the origin, and z is the coordinate along the symmetry axis of the nucleus. This expression compares the elongation along the symmetry axis to the average radius. A positive Q_0 corresponds to a prolate shape (rugby ball-like), while a negative Q_0 implies an oblate shape (pancake-like).

The intrinsic quadrupole moment is not directly measurable in the laboratory frame, but it connects to observable quantities like the spectroscopic quadrupole moment Q or the reduced electric quadrupole transition probability $B(E2)$. For a transition from a 0^+ to 2^+ state, the relation to Q_0 is:

$$B(E2; 0^+ \to 2^+) = \frac{5}{16\pi} Q_0^2 \tag{3}$$

This equation serves as the bridge between observed electromagnetic transitions and the intrinsic nuclear shape.

Later sections will show how measurements of gamma-ray emission intensities can be used to extract $B(E2)$ values, which are then inverted via this equation to determine Q_0 and, ultimately, the deformation parameter β_2.

1.3 Reduced Transition Probability $B(E2)$

The reduced transition probability $B(E2)$ quantifies the probability that a nucleus in an excited state will emit an electric quadrupole (E2) gamma-ray photon and transition to a lower energy state. In particular, the transition from a first excited 2^+ state to the ground 0^+ state in even-even nuclei is a standard probe of quadrupole collectivity. A larger $B(E2)$ value typically indicates greater collectivity and deformation.

Mathematically, the $B(E2)$ value is related to the square of the matrix element of the electric quadrupole operator $\hat{Q}_{2\mu}$:

$$B(E2; I_i \to I_f) = \frac{1}{2I_i + 1} \left| \langle I_f || \hat{Q}_2 || I_i \rangle \right|^2 \tag{4}$$

For the special case of $0^+ \to 2^+$ transitions in axially symmetric nuclei, this reduces to:

$$B(E2; 0^+ \to 2^+) = \frac{5}{16\pi} Q_0^2 \tag{5}$$

where Q_0 is the intrinsic quadrupole moment of the nucleus, this relationship allows us to extract internal structure information (e.g., deformation) from experimental gamma-ray measurements.

1.4 Key Relations for Extracting β_2

The quadrupole deformation parameter β_2 is essentially a classification of how non-spherical a specific isotope shape could be in reference to either being prolate ($\beta_2 > 0$) or oblate ($\beta_2 < 0$). Coulomb excitation processes measured β_2 by using Eqns. 4 and 6 in conjunction with each other. A more concrete derivation of reduced transition probabilities and quadrupole deformation parameters can be found via these citations [2, 3].

$$|\beta_2| = \frac{4\pi}{3ZeR_0^2} \sqrt{B(E2; 0^+_{\text{g.s.}} \to 2^+_1)}. \tag{6}$$

2 Experimental Methods

This chapter presents the experimental design and key measurements that enable the extraction of $B(E2)$ values through Coulomb excitation. The techniques described here form the foundation for the analysis procedures discussed in Chapter 3. The chapter can be summarized into three discussions: principles of Coulomb excitation and inverse kinematics; considerations in the Coulomb experiment; and examples of detector systems used and the technologies they represent. The experimental devices and methodologies help obtain gamma spectra that can be used to determine gamma yields and to obtain an angular correlation factor (W_γ), which is also needed to determine the coulex target cross-section and/or projectile cross-section. The angular correlation factor will be discussed in Chapter 3 along with other observables.

2.1 Principle of Coulomb Excitation

Coulomb excitation is a process in which a nucleus is excited by electromagnetic interaction with another nucleus (excitation can occur for either the target or the projectile, or both). To ensure that only the electromagnetic force, not the strong nuclear force, is involved, the beam energy is kept below the Coulomb barrier. This ”safe energy” condition guarantees that the excitation is purely electromagnetic. After excitation, the nucleus de-excites by emitting gamma rays, which carry information about its internal structure. The intensity (gamma energy spectra) of these gamma rays is directly linked to the $B(E2)$ transition probability, a quantity we aim to extract [4].

2.2 Inverse Kinematics

Because many nuclei of interest are unstable, the experiment uses inverse kinematics: a radioactive ion beam (RIB) is accelerated and directed onto a stable, heavy target. This setup ensures that the lighter target stays stationary primarily while the projectile continues forward, simplifying particle detection and improving angular coverage. Inverse kinematics is essential for studying exotic nuclei far from stability. An example thesis referenced for this paper states that the central projectile system being tested is ^{32}Mg or ^{30}Mg, and the target is composed of ^{60}Ni [2].

2.3 Target Considerations

In Coulex experiments, targets are chosen based on their stability, known electromagnetic properties (for example, ^{197}Au), and whether their interactions with the projectile would be purely electromagnetic. Targets for this paper [2] used for ^{58}Ni and ^{197}Au were chosen because of the already known transition properties of the gamma schemes, which, in an analysis like this, are helpful for calibrations that can be applied to energy spectra. Finally, other key details considered for a target include how well it can be Doppler-corrected (relativistic effects of a moving nucleus emitting gammas) and its angular resolution.

2.4 Experimental Observables

The primary observables in a Coulomb excitation experiment are the gamma-ray emissions from excited states and the scattering angles of the interacting nuclei. These observables provide the core input for determining excitation cross sections, which are essential for extracting $B(E2)$ values. Gamma-ray spectra help us identify the proper transitions of study and, essentially, the process of obtaining those yields and associating them with angular correlation factors (W^p_γ and W^t_γ).

These observables are combined into a cross-section ratio equation (see equation. 7) That links the projectile and target nuclei:

$$\sigma_{\text{CE}}^{p} = \frac{\epsilon_\gamma^t}{\epsilon_\gamma^p} \cdot \frac{b_\gamma^t}{b_\gamma^p} \cdot \frac{W_\gamma^t}{W_\gamma^p} \cdot \frac{N_\gamma^p}{N_\gamma^t} \cdot \sigma_{\text{CE}}^{t} \tag{7}$$

Here:

- $\epsilon_\gamma^{p,t}$ are the gamma-ray detection efficiencies,
- $b_\gamma^{p,t}$ are the gamma-ray branching ratios,
- $W_\gamma^{p,t}$ are the angular correlation (anisotropy) factors,
- $N_\gamma^{p,t}$ are the measured gamma-ray yields,
- σ_{CE}^{t} is the known excitation cross section of the target.

By measuring the gamma intensities and angles from both the projectile and the target, and referencing the known target excitation cross section, the $B(E2)$ value of the projectile can be determined through normalization.

2.5 Detector Systems

The detection systems used in Coulomb excitation experiments are carefully chosen to measure the key observables: gamma-ray energies and scattering angles. Two broad categories of detectors are typically employed:

Silicon Particle Detectors: These detectors are used to measure the energy and angle of scattered nuclei following the excitation. Typically arranged in a disk or ring geometry around the target, they provide angular resolution and enable particle identification. The primary role of silicon detectors is to determine which nucleus scattered (target or projectile), its trajectory, and its scattering angle. In some configurations, a ΔE-E telescope is used, where a thin detector (ΔE) sits in front of a thicker one (E). By analyzing how much energy a particle loses in each layer, one can identify the particle type, which is crucial for resolving interactions and separating background events.

High-Purity Germanium (HPGe) gamma-ray detectors: gamma-ray detectors, such as the MINIBALL array ([2]), are designed to record the energies and intensities of gamma rays emitted from excited nuclear states. Their main function is to build gamma-ray spectra, in which each peak corresponds to a transition from an excited state to a lower-energy level. These detectors have excellent energy resolution, which is essential for distinguishing closely spaced gamma lines. Their high segmentation also allows for position sensitivity and Doppler correction—important because the emitting nuclei move rapidly when they emit gamma rays. Without proper Doppler correction, the gamma-ray peaks would be smeared, reducing the precision of energy measurements.

Together, these detector systems ensure comprehensive coverage of the reaction observables. Particle detectors define the scattering geometry, enabling the calculation of angular-correlation factors.

In contrast, gamma-ray detectors provide the spectral information needed to identify excited states and quantify their transition strengths. These measurements form the basis for extracting $B(E2)$ values using data analysis tools such as GOSIA.

In summary, Chapter 2 outlines the experimental techniques and measurements that form the input for analysis. The particle detectors provide angular data and particle identification, while the gamma detectors supply energy and intensity information. These data are then corrected, normalized, and analyzed in Chapter 3 using the GOSIA code to extract matrix elements and ultimately determine $B(E2)$ and β_2 values.

3 Methodology for Extracting $B(E2)$ and β_2

This chapter outlines the procedure used to extract reduced transition probabilities $B(E2)$ and quadrupole deformation parameters β_2 from Coulomb excitation experiments (described in chapter 2), with a focus on the analysis pipeline used in experiments like those described in [2]. The overall strategy involves gamma-ray identification, efficiency corrections, extraction of target and projectile yields, angular-distribution analysis (W's are calculated), normalization, matrix-element extraction via GOSIA, and the final computation of $B(E2)$ and β_2.

3.1 Gamma-Ray Identification

Gamma-ray spectra from the MINIBALL array are used to identify transitions from excited states. Doppler correction is applied on an event-by-event basis using the known kinematics of scattered ions. Peaks in the gamma spectra are matched to known energy levels, enabling yield extraction for specific transitions. An example of gamma spectra and yield extraction is observed through Figure 3.

3.2 Efficiency Corrections

Detector efficiencies $\epsilon_\gamma(E)$ are determined using GEANT4 simulations of the MINIBALL array for each gamma energy and angular segment. Both absolute and relative efficiency curves are generated, accounting for the effects of cluster add-back (i.e., adding coincident signals from nearby detector regions). These efficiencies are then used to correct the measured gamma yields (see Figure 4).

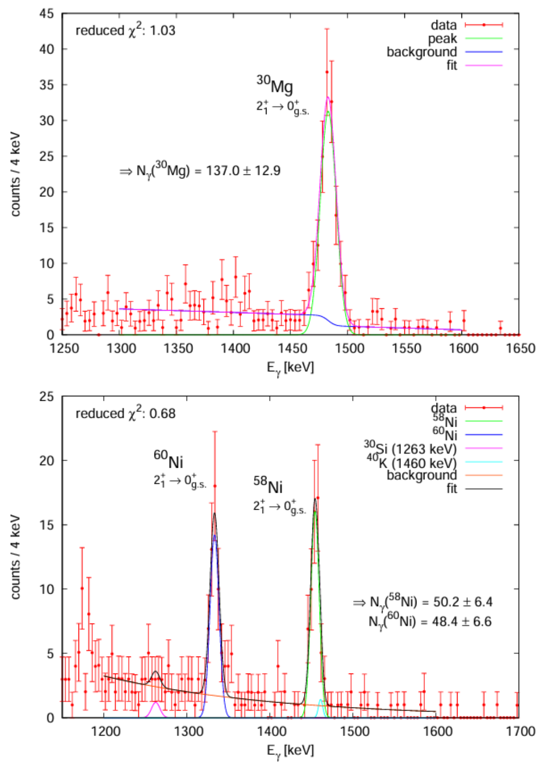


Figure 3: This example figure shows what gamma spectra should look like in terms of output of transitions, fitting, and using fitting information to extract target and projectile yields. This example specifically deals with gammas from magnesium isotopes as projectiles and nickel isotopes as targets. Image from [2].

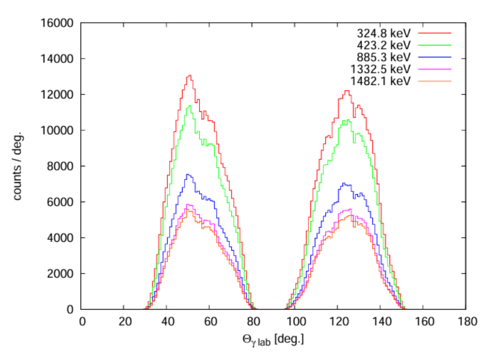


Figure 4: Geant-4 efficiency determination from simulations of detectors of interest from [2]. An important note is that gammas is being modeled as different energy transitions.

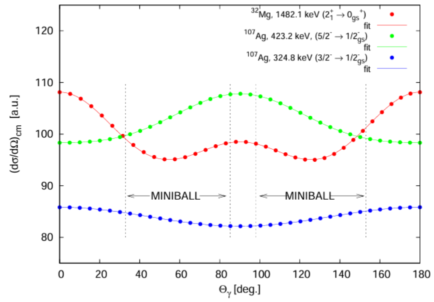


Figure 5: Differential cross-section of coulex interactions modeled as an angular distribution that's fitted for target and projectile gamma information. The ratio of distributions is what leads to the import observable in equation 7. Image taken from [2].

3.3 Angular Yield Sorting and Distributions

Corrected gamma yields are binned as a function of scattering angle Θ_{lab}, transformed to the center-of-mass frame using average recoil velocities:

$$\cos(\Theta_{\gamma,\text{cm}}) = \frac{\cos(\Theta_{\gamma,\text{lab}}) - \langle\beta\rangle}{1 - \langle\beta\rangle \cos(\Theta_{\gamma,\text{lab}})} \tag{8}$$

The resulting angular distributions are fit with Legendre polynomials to model anisotropy. the division between target Figure 5 allows one to extract the ratio ($\frac{W^t_\gamma}{W^p_\gamma}$ of angular correction factors of target and projectile (part of equation 7)

3.4 Normalization Procedures

Relative cross sections are normalized using either:

- Excitation of a well-known target transition (e.g., ^{107}Ag or ^{197}Au) with known $B(E2)$.
- Rutherford scattering yields in silicon detectors.

The cross-section ratio is given by: equation 7.

3.5 GOSIA Analysis and extraction of $B(E2)$

The GOSIA code performs a least-squares fit of measured gamma-ray yields across angular bins. The program simulates Coulomb excitation and de-excitation using a coupled-channels approach, accounting for geometry, multipole mixing, detector responses, and other details to match experimental observables. GOSIA varies matrix elements until simulated yields match experimental values [5]. One should note that the matching of the simulation gamma yield to the data, as indicated in equation 7, occurs when B(E2) is extracted, because one varies B(E2) from the matrix elements until the simulation and data match.

3.6 Calculation of β_2

Finally, the deformation parameter β_2 is computed from the usage of equation 6. This pipeline—from raw spectra to final β_2 values—enables a complete structural analysis of exotic nuclei using Coulomb excitation.

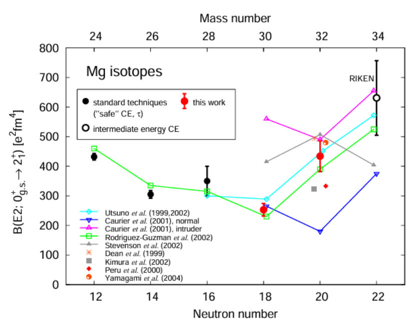
4 Selected Results from Literature

4.1 ^{30}Mg and ^{32}Mg

This section summarizes experimental results for neutron-rich magnesium isotopes, particularly ^{30}Mg and ^{32}Mg, obtained via Coulomb excitation. These results provide key insights into the evolution of quadrupole collectivity and the breakdown of shell structure near the so-called "island of inversion." The extracted $B(E2)$ values and deformation parameters β_2 are critical benchmarks for both nuclear structure theory and the performance of experimental tools such as GOSIA. The interpretation of the magnesium isotopic results indicates the presence of prolatenes, as shown in Figure 6, and the B(E2) values provide some indication that the shell model is beginning to fail for these isotopes.

Isotope	$B(E2;\ 0^+_{\text{g.s.}} \to 2^+_1)\ [e^2\text{fm}^4]$	$t_{1/2}(2^+_1)$ [ps]	$\lvert\beta_2\rvert$
^{30}Mg	253 (21)	1.55 (13)	0.400 (17)
^{32}Mg	434 (52)	11.9 (14)	0.501 (30)

(a) Summary table showing extracted values of $B(E2)$, β_2, and excited-state lifetimes for ^{30}Mg and ^{32}Mg. Notably, ^{32}Mg shows a larger $B(E2)$ and β_2, indicating collectivity and deformation.



(b) $B(E2; 0^+ \to 2^+)$ systematics for Mg isotopes. Red points represent this work using "safe" Coulomb excitation. The trend shows a dramatic increase in $B(E2)$ at $N = 20$ for ^{32}Mg.

Figure 6: Experimental determination and systematics of $B(E2)$ and β_2 for neutron-rich magnesium isotopes. Together, the figures demonstrate enhanced collectivity at $N = 20$, placing ^{32}Mg. The $B(E2)$ values due indicate post shell-model predictions

5 Conclusion and Outlook

This paper presented a compact overview of how to extract the quadrupole deformation parameter β_2 through measurements of the reduced transition probability $B(E2)$ using Coulomb excitation reactions. The theoretical foundation established how deformation arises from collective nuclear behavior and how $B(E2)$ and β_2 are linked through intrinsic quadrupole moments.

Coulomb excitation was introduced as a clean electromagnetic probe, ideal for measuring transition strengths in unstable nuclei. The relationship between observed gamma-ray yields and structural parameters was formalized, and the safe-energy condition required for pure Coulomb excitation was emphasized.

Using [2] as a detailed case study, Chapter 2 outlined the experimental methodology, including inverse kinematics, target selection, and the use of segmented gamma-ray detectors (e.g., MINIBALL) and silicon particle detectors. These tools enable accurate measurement of gamma spectra and angular distributions, which are essential inputs for the analysis.

Chapter 3 described the data analysis pipeline: identifying de-excitation gamma peaks, applying efficiency corrections, performing angular binning, and calculating gamma yields. These inputs are used in GOSIA simulations, which iteratively adjust transition-matrix elements until simulated gamma yields match experimental data. From these fits, $B(E2)$ values are extracted and then converted to β_2.

Chapter 4 demonstrated this process with results for magnesium isotopes. The extracted values, such as $\beta_2 \approx 0.5$ for ^{32}Mg and ^{30}Mg, confirm a strongly prolate shape consistent with the collective behavior post shell model behavior. These results underscore how Coulomb excitation and gamma-ray analysis provide sensitive probes of nuclear deformation.

Specific approaches for measuring β_2 could come from other techniques, such as collision experiments that address structure questions in their own ways [6]. Also probing other nuclei and using coulex experiments to determine whether interesting deformation information can be obtained, such as in [7].

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